Motasem ElGamel

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Professional appointments

Tarbutton Postdoctoral Fellow **2026 - Present**
Theory and Modeling of Living Systems Initiative, Emory University *Atlanta, GA*

Education

University of Pittsburgh **2019 - 2025**
Physics - Ph.D. *Pittsburgh, PA*
Advisor: Prof. Andrew Mugler.
Dissertation: Microbial Dynamics at the Cellular and Population Levels.

University of Pittsburgh **2019 - 2021**
Physics - M.Sc. *Pittsburgh, PA*

University of Science and Technology at Zewail City **2015 - 2019**
Physics - B.Sc. (Graduated Magna Cum Laude) *Giza, Egypt*

Publications

*Equal contribution

5. K. P. Ramachandran, **M. ElGamel**, F. Jafarpour, and A. Mugler. Optimal information transmission in a sequential model for cell division. *arXiv preprint arXiv:2605.03173*, 2026.
4. **M. ElGamel**, L. Ribaudo, and A. Mugler. General mechanism for concentration-based cell size control. *Physical Review Research*, 8(3):033012, 2026.
3. S. Sayin*, **M. ElGamel***, B. Rosener, M. Brehm, A. Mugler, and A. Mitchell. Bacterial population dynamics during colonization of solid tumors. *Molecular Systems Biology*, 22(3):412–434, 2026.
2. **M. ElGamel** and A. Mugler. Effects of molecular noise on cell size control. *Phys. Rev. Lett.*, 132:098403, 2024 - Press release (link).
1. **M. ElGamel**, H. Vashistha, H. Salman, and A. Mugler. Multigenerational memory in bacterial size control. *Physical Review E*, 108(3):L032401, 2023.

Workshops and Schools

Physics and Computation in Immunology 6th Workshop and Symposium **September 2026**
Yale University *New Haven, CT*

NITMB and SMTPB Modeling and Theory in Population Biology Workshop **June 2025**
National Institute of Theory and Mathematics in Biology *Chicago, IL*

- Competitively selected for a fully funded position.

NITMB Ecological Dynamics of Microbial Communities: New Approaches Workshop **May 2024**
National Institute of Theory and Mathematics in Biology *Chicago, IL*

Les Houches Theoretical Biophysics Summer School 2023 **July 2023**
École de physique des Houches *Les Houches, France*

- A month-long school aimed at elucidating the function and principles of biological systems, from cells to ecosystems, in light of the laws of physics.
- Worked on assessing selection and coevolution in protein sequences as part of a group project.

Talks and Presentations

1. Invited talks

The University of Iowa Math Bio Seminar (virtual) **March 2026**
Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria"

Quantitative Biology Institute, Yale University **May 2025**
Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *New Haven, CT*

Department of Physics, Emory University **April 2025**
Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Atlanta, GA*

The National Institute for Theory and Mathematics in Biology **December 2024**
Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Chicago, IL*

Group meeting of Prof. Michael Stumpf (virtual) **November 2024**
Invited talk: "Effects of molecular noise on cell size control" *The University of Melbourne, Australia*

SIAM Conference on the Life Sciences 2024 **June 2024**
Invited talk: "Clone size statistics of tumor-inhabiting bacteria" *Portland, OR*

2. Selected talks

QBio Chicago 2026 **July 2026**
Selected talk: "Spatial structure promotes microbial resilience to environmental fluctuations" *Chicago, IL*

NITMB and SMTPB Modeling and Theory in Population Biology Workshop **June 2025**
Selected talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Chicago, IL*

Gordon Research Conference and Seminar **January 2025**
Selected talk and poster presentation: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Ventura, CA*

APS March Meeting 2024 **March 2024**
Selected talk for DBIO Early Career Prize Session: "Clone size statistics of tumor-inhabiting bacteria" *Minneapolis, MN*

Gordon Research Conference and Seminar **January 2023**
Selected talk and poster presentation: "Effects of molecular noise on cell size control" *Ventura, CA*

NSF Center for the Physics of Biological Function - Physics of Life Symposium **November 2022**
Selected talk: "Theory of cell size homeostasis" *New York City, NY*

The 2nd Biology for Physics Conference **July 2022**
Selected talk: "Multigenerational memory in bacterial size control" *Barcelona, Spain*

3. Contributed, seminar, and poster presentations

APS Global Physics Summit 2027 **April 2027**
Contributed talk: "Evolutionary consequences of extrinsic stochasticity in within-host viral populations" *Atlanta, GA*

Physics and Computation in Immunology 6th Workshop and Symposium **September 2026**
Poster presentation: "A theory of immune-driven type 2 diabetes" *New Haven, CT*

Population Biology, Ecology, and Evolution Seminar, Emory University **April 2026**
Seminar: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Atlanta, GA*

APS Global Physics Summit 2026 **March 2026**
Contributed talk: "Spatial structure promotes microbial resilience to environmental fluctuations" *Denver, CO*

Pitt Biological Physics Day **September 2025**
Poster presentation: "General mechanism for concentration-based cell size control" *Pittsburgh, PA*

APS Global Physics Summit 2025 **March 2025**
Contributed talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Anaheim, CA*

Pitt Biological Physics Day **September 2024**
Poster presentation: "Population dynamics and universal statistics of tumor-inhabiting bacteria" *Pittsburgh, PA*

NITMB Ecological Dynamics of Microbial Communities: New Approaches Workshop **May 2024**
Lightning talk: "Clone size statistics of tumor-inhabiting bacteria" *Chicago, IL*

University of Pittsburgh Grad Expo 2024 **March 2024**
Contributed talk: "Clone size statistics of tumor-inhabiting bacteria" *Pittsburgh, PA*

Applied Math Seminar, Department of Mathematics, University of Pittsburgh **March 2024**
Seminar: "Growth Dynamics of Tumor-Inhabiting Bacteria" *Pittsburgh, PA*

Les Houches Theoretical Biophysics Summer School 2023 **July 2023**
Poster presentation: "Theory of noisy population growth in bacteria" *Les Houches, France*

Physics of Life 2023 Conference **March 2023**
Poster presentation: "Effects of molecular noise on cell size control" *Harrogate, UK*

APS March Meeting 2023 **March 2023**
Contributed talk: "Adder minimizes cell size noise in bacteria" *Las Vegas, NV*

APS March Meeting 2022 **March 2022**
Contributed talk: "Multigenerational memory in cell size homeostasis" *Chicago, IL*

Fellowships and Awards

Tarbutton Postdoctoral Fellowship **2026 -**
Theory and Modeling of Living Systems Initiative, Emory University

APS FIP Distinguished Student Travel Award **2026**
American Physical Society (APS) and Forum on International Physics (FIP)

Selected Speaker at the DBIO Early Career Prize Session **2024**
American Physical Society (APS) March Meeting 2024

Outstanding Presentation Award Pitt Grad Expo 2024 **2024**
Dietrich School of Arts and Sciences, University of Pittsburgh
- Awarded to the talk: "Clone size statistics of tumor-inhabiting bacteria".

Andrew W. Mellon Predoctoral Fellowship **2023 - 2024**
University of Pittsburgh
- Awarded school-wide to doctoral students of exceptional promise and ability across the disciplines based on a research proposal. Consists of funding for one year.

Best Poster Award at the Physics of Life 2023 Conference, Harrogate, UK **2023**
- Awarded to the poster presentation: "Effects of molecular noise on cell size control".

Pitt Graduate and Professional Student Government (GPSG) Travel Grant **2023**
University of Pittsburgh

Thomas-Lain Fund Scholarship Essay Competition **2023**
Department of Physics and Astronomy, University of Pittsburgh

Mentorship

Research Mentor
Department of Physics and Astronomy, University of Pittsburgh
- Mentored undergraduate students Jianan Zhao and Lucas Ribaudo* in research.

* Awarded the Emil Sanielevici Undergraduate Research Scholarship for research on concentration-based cell size control in yeast.

Peer-Mentor in ZC Alumni Mentorship Program
Zewail City Alumni Association

- Mentored one undergraduate student.

Graduate Student Mentor

Department of Physics and Astronomy, University of Pittsburgh

- Mentored two incoming graduate students.

Teaching

Graduate Teaching Assistant

University of Pittsburgh

- PHYS 0110: Introduction to Physics 1 - PHYS 0111: Introduction to Physics 2 - PHYS 0174: Basic Physics for Science and Engineering 1.
- Lead recitations, held office hours, prepared quizzes, graded exams and wrote up homework solutions.

Undergraduate Junior Teaching Assistant

University of Science and Technology at Zewail City

- PHYS 101: Introduction to Physics 1 - PEU 438: Compact Objects.

Outreach and Service

Journal Reviewer

- Reviewed articles for *Physical Review Letters*, *Physical Review E*, *Physical Review X Life* (with advisor), *Physical Review X* (with advisor), *Nature Communications Physics*, and *Springer's Bulletin of Mathematical Biology.*

Co-chair of the Stochastic Physics in Biology Gordon Research Seminar **January 2027**

- Organizing the next Stochastic Physics in Biology Gordon Research Seminar in 2027. *Ventura, CA*

Lecturer at the 1st Zewail City Alumni Physics Winter School (virtual) **January 2025**

University of Science and Technology at Zewail City *Zewail City, Egypt*

- *Lecture: "How Do Cells Navigate a Noisy Life?".*
- *General introductory lecture on biological physics.*

Zewail City Science Festival **2016 - 2017**

University of Science and Technology at Zewail City *Zewail City, Egypt*

- Event that aims to deliver scientific concepts in a simplified manner to the public and spread the culture of science.
- Participated as an organizer, wrote a scientific talk for the public audience, and gave a mathematics talk.

Skills

Programming Languages: C++, Python, Matlab, Mathematica, LabVIEW.
Computational: Stochastic Simulation (Gillespie) Algorithm, Monte Carlo Simulations, Stochastic simulations, Data Analysis.
Modeling: Dynamical Systems, Stochastic Processes, Statistical Physics, Complex Analysis.
Certificates: Deep Learning Specialization (link, 5 courses: Networks and Deep Learning - Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization - Structuring Machine Learning Projects - Convolutional Neural Networks - Sequence Models).